Neighborhood Effects on Educational Attainment by Disability Status

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Abstract

Children who grow up in higher-opportunity places attain higher levels of education, but little is known about whether these gains extend to students with disabilities. Using administrative data on 1.2 million Texas students and variation in the grade at which they move across districts, I estimate how years of exposure to a higher-opportunity district affect educational attainment by disability status. Longer exposure raises high school completion, college attendance, and college completion, but the timing of these gains differs by disability status. For students without disabilities, the gains come mainly from moves in elementary school, and moves after grade 5 have no detectable effect. For students with disabilities, moves after grade 5 still raise attainment. A student with a disability who moves in grade 6 rather than grade 12 to a district with a one standard deviation higher college attendance rate is 3.4 percentage points (about 10%) more likely to attend college.

Keywords: neighborhood effects, childhood exposure, educational attainment, disability

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1. Introduction

Where children grow up shapes how much education they attain and how much they earn as adults. Children who move to better neighborhoods, especially at younger ages, attain more education, earn more, and have fewer arrests for violent crimes (Chetty et al. 2016; Chetty and Hendren 2018a; Chyn 2018; Laliberté 2021; see Chyn and Katz 2021 and Katz 2026 for reviews). These findings suggest that policy can improve children's long-run outcomes by changing the places where they grow up. Housing mobility programs help families with vouchers move to higher-opportunity neighborhoods, and place-based programs invest in the neighborhoods where children already live (Katz 2026). Because the gains grow with years of exposure, researchers recommend directing these programs to families with young children rather than leaving families on waitlists while their children age (Chetty et al. 2016; Chyn and Katz 2021).

Much less is known about which children gain the most from better places, although estimates for specific groups of children are a priority for research that aims to guide families and policymakers (Chetty and Hendren 2018b). Studies find that the effects of places differ by gender, family income, and immigrant origin (Chetty and Hendren 2018a, 2018b; Deutscher 2020; Borgen et al. 2025). However, few studies have examined students with disabilities (SWDs). Recent studies of how housing and schools affect children's outcomes include special education status as a control variable but do not estimate effects separately for SWDs (Han and Schwartz 2025; Mbekeani et al. 2026).

This omission matters for policy because SWDs are a large group and attain less education than other students. In the 2022–23 school year, 7.5 million U.S. students ages 3 to 21 received special education services, the equivalent of 15% of all public school students (National Center for Education Statistics 2024a). Nationally, 71% of SWDs graduated from high school within four

years in the 2021–22 school year, compared with 87% of all students (National Center for Education Statistics 2024b). In Texas, SWDs are much less likely than other students to attend and complete college. Among students in grade 5 between 2000 and 2005, 32.7% of SWDs attended college, compared with 56.2% of students without disabilities, and 6.3% completed college, compared with 19.7% (Ballis and Heath 2021).

There are reasons to expect places to affect SWDs differently from other students, and at different ages. First, SWDs rely more than other students on services available where they grow up, such as special education. Second, many decisions that shape the attainment of SWDs, such as transition planning, occur in middle and high school, so places may continue to matter in adolescence. Third, moving can disrupt the services and relationships that SWDs rely on, which could reduce their gains from a move. Because these forces point in different directions, whether places matter more or less for SWDs, and at what ages, is an empirical question.

I examine this question using administrative records on 1.2 million students who entered kindergarten in Texas public schools from 1994 to 1999, linked to records of enrollment and degrees at Texas colleges. I study three outcomes, high school completion, college attendance, and college completion. Following Chetty and Hendren (2018a), I measure the opportunity of each school district by the attainment of students who stay in it from kindergarten through grade 12. I then compare students from the same origin district and cohort who move once, at different grades, to districts with different levels of opportunity, and estimate how much each additional year in a higher-opportunity district raises attainment, separately for SWDs and students without disabilities. The identifying assumption is that the differences between families who move to higher- and lower-opportunity districts do not depend on the grade at which their child moves. I test this assumption in Section 6.

I find three main results. First, places shape educational attainment. Each additional year in a district where more students attend college closes about 1.1% of the gap in college attendance between the origin and destination districts, and the rates for high school completion and college completion are similar. Second, for college attendance, but not for the other outcomes, the gains are twice as large for SWDs as for students without disabilities (2.4% compared with 1.2% of the gap per year). Third, moves after grade 5 have no detectable effect for students without disabilities but still raise attainment for SWDs. An SWD who moves in grade 6 rather than grade 12 to a district with one standard deviation higher college attendance is 3.4 percentage points more likely to attend college, about 10% of the mean for SWDs. Placebo tests with test scores measured before the move show little evidence that families choose the timing of their moves based on their children's achievement, and the larger effect on college attendance for SWDs who move after grade 5 remains when I compare only students who make the same move in the same cohort.

This paper makes three contributions. First, it adds to research on who benefits from better places. Studies of this question have compared children by gender, family income, and immigrant origin. I extend this work to SWDs and show that the effects of places differ by disability status, especially for moves in middle and high school. Second, it adds to the evidence on the timing of exposure. Studies disagree about whether moves in adolescence benefit children (Chetty et al. 2016; Chetty and Hendren 2018a; Deutscher 2020). Within a single design and setting, I show that the answer differs across groups of students. Third, it adds to research on the education of SWDs, which has mostly asked whether receiving special education services improves achievement and attainment (Ballis and Heath 2021; Schwartz et al. 2021). I show that where SWDs grow up also shapes their attainment.

The results show that educational opportunity for SWDs is also tied to place, which has implications for housing and education policy. Housing mobility programs often focus on families with young children because the gains from moving grow with years of exposure (Chyn and Katz 2021). For SWDs, however, moves in middle and high school still raise attainment, so age limits may exclude families for whom a later move would still pay off. Improving the places where SWDs already live, including the services available in middle and high school, could complement helping families move. Section 7 discusses these implications and the limits of the evidence.

2. Background

2.1 Neighborhood Effects and the Timing of Exposure

Children who grow up in different places have very different outcomes. These differences could arise for two reasons (Katz 2026). Places may have causal effects, so that moving a given child to a different place would change the child's outcomes. Alternatively, the differences may reflect the types of families who live in each place.

Studies separate the two using experiments and the timing of family moves. In the Moving to Opportunity experiment, families in public housing were randomly offered vouchers to move to low-poverty neighborhoods. For children younger than 13 at the time of the offer, the offer raised college attendance at ages 18 to 20 by 2.5 percentage points, or 16% (Chetty et al. 2016). Chetty and Hendren (2018a) compare children whose families move across commuting zones at different ages. Children's incomes converge to those of children who grow up in the destination at about 4% per year of exposure, and their college attendance converges at a similar rate. Studies in other settings find similar patterns. In Montreal, movers' educational attainment converges at about 4.5%

per year, and access to better schools accounts for 50% to 70% of this effect (Laliberté 2021). In Texas, more years in counties where more non-movers earn STEM degrees raise the probability that movers earn one (Kim et al. 2026). Whether places still matter in adolescence is less settled, and the answer matters for policy. If places affect children mainly at certain ages, policy should improve children's environments at those ages, but if each year matters, policy should sustain improvements throughout childhood (Chetty and Hendren 2018a). In Moving to Opportunity, the offer of a low-poverty voucher lowered college attendance at ages 18 to 20 by 4.3 percentage points for children aged 13 to 18, and the authors suggest the disruption of moving to a very different environment in adolescence as one potential explanation (Chetty et al. 2016). Among children displaced by public housing demolitions in Chicago, those displaced at ages 7 to 12 were 5.1 percentage points less likely to drop out of high school, while those displaced at ages 13 to 18 showed no detectable change in schooling (Chyn 2018). Chetty and Hendren (2018a), in contrast, find that each year of childhood exposure matters roughly equally and conclude that better neighborhoods can benefit children even in adolescence. Because their estimates compare families who all move, they net out the fixed costs of moving that the experiment does not. Other studies point to adolescence as a period of particular sensitivity. In Australia, children's incomes converge at about 1.1% per year of exposure before age 11 and 4.2% per year after (Deutscher 2020), and in the United States, neighborhoods appear to have slightly larger effects in adolescence than in the earliest years of childhood (Chetty et al. 2026). Using the same Texas data as this paper, Kim et al. (2026) find that each year of exposure in grades 8 to 11 has a larger effect on STEM degree attainment than a year in grades 1 to 7, which they interpret as consistent with the role of high school in course taking and career plans.

The effects of places also differ across groups of children. Places matter more for boys than for girls, especially in low-income families, and the places that produce the best outcomes for children in low-income families are not always those that produce the best outcomes for children in high-income families (Chetty and Hendren 2018b). In Norway, children whose parents come from Asia, Sub-Saharan Africa, or the Middle East and North Africa are less affected by neighborhood disadvantage in their grade 5 achievement than children of native-born parents and children of immigrants from other regions (Borgen et al. 2025). In Texas, exposure effects on STEM degree attainment are positive for every racial and economic group but vary in size across groups (Kim et al. 2026). None of these studies examines whether the effects of places differ by disability status.

2.2 Why Neighborhood Effects Might Differ by Disability Status

The effects of places on SWDs could be larger or smaller than for other students, and they could arise at different ages. Under the Individuals with Disabilities Education Act (IDEA), public schools must provide special education to eligible children, and for each child a team that includes the child's parents and teachers writes an Individualized Education Program (IEP) that states the services the child will receive. These services are delivered locally. Three features of this system suggest why the effects of places could differ for SWDs.

First, SWDs rely more than other students on services available where they grow up, and these services vary across places in two ways. Districts and states implement federal special education law differently, so the placements available to SWDs can differ with where they live (Kluth et al. 2007). The same policy can also have different consequences depending on local

resources. When a statewide policy limited access to special education in Texas, high school completion and college enrollment among students in special education declined significantly in less wealthy and lower-achieving districts but not in wealthier or higher-achieving ones (Ballis and Heath 2021). SWDs also start from lower attainment, which leaves more room for places to raise it. These features suggest larger effects of places for SWDs.

Second, moving may disrupt the services and relationships that SWDs rely on. When a student with an IEP moves to another district in the same state during the school year, the new district provides comparable services until it either adopts the old IEP or develops a new one (34 C.F.R. § 300.323(e)). Interviews with families of children with disabilities show that this transition can be difficult, with several families facing new struggles with their children's inclusion in the years after moving across districts (Kluth et al. 2007). Moves also weaken adolescents' friendship networks (South et al. 2007), and these losses may be larger for SWDs. Adolescents with learning disabilities report more loneliness than other adolescents (Al-Yagon 2012), and among high school students with disabilities, relationships with teachers, which a move also interrupts, are associated with school bonding and adjustment (Pham and Murray 2016). If these disruptions are costly, SWDs could gain less from moving to a higher-opportunity district, especially when they move in later grades and have fewer years to recover.

Third, places may continue to matter for SWDs in middle and high school, even though the evidence that moves at these ages benefit other children is mixed (Section 2.1). Most SWDs identified by grade 5 remain in special education into high school. In Texas, more than 70% of students in special education in grade 5 still receive services in grade 9 (Ballis and Heath 2021). During these grades, schools make decisions that can set an SWD's path toward college. As graduation approaches, IEPs turn to transition plans that guide SWDs toward college or toward

employment and career and technical courses (Ballis and Heath 2021). If districts differ in how they shape these plans, a move in middle or high school can change an SWD's path after high school. Because these plans concern what students do after high school, the effects of such moves may show up most clearly in college attendance.

These features do not all point in the same direction. Greater reliance on local services and the importance of decisions in middle and high school suggest larger and later effects for SWDs, while the disruption of services and relationships suggests smaller effects, especially for late moves. Whether places matter more or less for SWDs, and at what ages, is therefore an empirical question. Sections 5 and 6 examine it by comparing exposure effects for SWDs and students without disabilities, separately for students who move before and after grade 5.

3. Data

3.1 Data Source and Sample

I use restricted-access administrative records from the Texas Schools Project, housed at the Education Research Center at the University of Texas at Dallas. The data link two sources. The first is the Texas Education Agency (TEA) records, which cover every student enrolled in a Texas public school from kindergarten through grade 12. For each school year, these records report the student's school district, demographic characteristics, eligibility for free or reduced-price lunch, English learner status, special education status and primary disability, and scores on state standardized tests. The second is the Texas Higher Education Coordinating Board (THECB) records, which cover enrollment and degree receipt at all Texas colleges and universities.

The sample begins with 1,244,591 students who enter kindergarten in a Texas public school in the fall of 1994 through 1999. I classify each student by the number of times they change school districts between kindergarten and grade 12. Non-movers never change districts (679,810 students, 54.6%), one-time movers change districts once (347,586, 27.9%), and multi-movers change districts more than once (217,195, 17.5%) (Table A1).

I estimate exposure effects using one-time movers, following Chetty and Hendren (2018a, 2018b). Each one-time mover has a single origin and a single destination, so years of exposure to the destination are well defined. Multi-movers pass through a sequence of districts, and their exposure to any one district cannot be summarized by a single number. I define the grade at move as the first grade in which a student is enrolled in the destination district. Exposure duration is 13 minus the grade at move. A student who moves in grade 1 spends 12 years in the destination district, and a student who moves in grade 12 spends one year. The mean exposure duration is 7.55 years (Table A2).

Table 1 presents summary statistics for one-time movers by disability status. Of the 347,586 one-time movers, 58,095 (16.7%) are SWDs. Compared with students without disabilities, SWDs are more likely to be male, more likely to be eligible for free or reduced-price lunch, and more likely to be English learners. They also score lower on grade 3 reading and math tests, conditional on having a score in the data. SWDs also have lower educational attainment, and the gap opens at the transition from high school to college.[2] High school completion is 84.8% for

[2] High school completion is an indicator for graduating from a Texas public high school. College attendance is an indicator for enrolling in a Texas college or university within four years of expected high school graduation. College completion is an indicator for earning a postsecondary degree, excluding certificates, within six years of expected high school graduation.

SWDs compared with 86.6% for students without disabilities, but college attendance is 35.3% compared with 60.3%, and college completion is 5.1% compared with 17.6%.

Table 1. Summary Statistics for One-Time Movers by Disability Status

	Overall	Without disabilities	With disabilities
Students	347,586	289,491	58,095
Outcomes			
High school completion	0.863	0.866	0.848
College attendance	0.557	0.603	0.353
College completion	0.154	0.176	0.051
Race and ethnicity			
Hispanic	0.431	0.433	0.422
White	0.376	0.377	0.372
Black	0.162	0.157	0.190
Asian	0.024	0.027	0.010
Native American	0.004	0.004	0.004
Two or more races	0.003	0.002	0.003
Sex			
Female	0.473	0.500	0.339
Male	0.527	0.500	0.661
Socioeconomic status			
Free or reduced-price lunch	0.456	0.440	0.537
English learner	0.035	0.024	0.090
Baseline test scores (SD)			
Grade 3 reading score	-0.089	0.006	-1.112
Grade 3 math score	-0.094	-0.003	-0.926
Disability type			
Learning disability	0.107		0.639
Speech impairment	0.020		0.121
Other health impairment	0.017		0.103
Emotional disturbance	0.009		0.055
Intellectual disability	0.007		0.043
Auditory or visual impairment	0.003		0.015
Autism	0.002		0.011
Orthopedic impairment	0.002		0.009
Traumatic brain injury	0.0004		0.002

Notes: The sample is one-time movers in the 1994 to 1999 kindergarten cohorts. The first row reports the number of students. Test score rows report means, and all other rows report shares. Grade 3 test scores are standardized and are observed for 89% of students without disabilities and 41% of students with disabilities. Disability types are shares of all students in the Overall column and shares of students with disabilities in the last column. Category names follow current terminology; intellectual disability appears as mental retardation in the administrative records. Auditory or visual impairment includes deaf-blindness, which is combined with these categories to protect student privacy. Estimation samples in later tables are smaller because of masked district opportunity, missing outcomes, and the removal of singleton fixed-effect cells.

3.2 Measuring Disability Status

I identify SWDs using school records of disability identification. TEA records whether each student receives special education services and the student's primary disability in every school year, and I classify students who receive special education services as SWDs. Disability status in this paper therefore refers to school identification, not to a medical diagnosis or to disability in adulthood.

As in Ballis and Heath (2021), who use records from the Texas Schools Project, I classify each student as an SWD or a student without disabilities based on disability status in grade 5 for the primary analysis. Many students with disabilities are not identified until the later elementary grades, so an earlier measure would classify them as students without disabilities. During the period I study, many schools did not assess children for learning disabilities before grade 3, an approach that the President's Commission on Excellence in Special Education (2002, 25–26) described as a "wait to fail" model. In national survey data on children who entered kindergarten in 1998, the same year as one of my cohorts, the share whose parents report a professional diagnosis of a learning problem rises from 2.6% in kindergarten to 13.2% in grade 5 (Dhuey and Lipscomb 2010). Using the same survey, Morgan et al. (2015) find that the rate of new learning disability identification peaks in grade 5. The data used in this paper show the same pattern. Among one-time movers, Table B2 shows that 81.3% of SWDs in grade 5 are not identified in kindergarten, and Table B1 shows that the share identified peaks at 16.7% in grades 5 and 6 and declines slowly afterward (see also Ballis and Heath 2021).[3]

[3] Two checks address remaining concerns about this measure. First, for students who move before grade 5, identification in grade 5 occurs after the move and could respond to the destination district, so Table 3 compares SWDs and students without disabilities among students who move in grades 6 to 12, whose status is determined before the move. Second, Tables B4 and B5 measure disability status in each of grades 1 to 4 instead of grade 5. Sections 4.3, 5.3, and 6.3 discuss both checks.

Measuring status in grade 5 also places it before a major change in Texas special education. In the 2004–05 school year, the Texas Education Agency set a target of 8.5% for each district's special education enrollment. Over the next ten years, the share of Texas students in special education fell from 13% to 8.5%, and in 2018 the U.S. Department of Education determined that the target violated federal law (Ballis and Heath 2021). The students I study reached grade 5 between the 1999–2000 and 2004–05 school years. Because special education status is recorded in October and districts received their first reports under the target in December 2004, their status in grade 5 was recorded before districts responded to the target (Ballis and Heath 2021).

3.3 Measuring District Opportunity

I measure the educational opportunity of each school district by the mean outcome of its non-movers in each kindergarten cohort, separately for each outcome, following Chetty and Hendren (2018a). I refer to this measure as district opportunity. Non-movers spend kindergarten through grade 12 in a single district, so their mean outcome summarizes the attainment of children who grow up there. Figure 1 maps college attendance among non-movers in each district, pooling the six cohorts.[4] College attendance is below 56.5% in the bottom quarter of districts and above 70.8% in the top quarter, a gap of more than 14 percentage points. Figure A1 shows similar variation in high school completion and college completion. Children's educational attainment differs widely by where in Texas they grow up.

[4] When a district has fewer than 20 non-movers in a cohort, I mask its opportunity for that cohort to protect student privacy. On the map, districts with fewer than 20 non-movers are shown in gray.

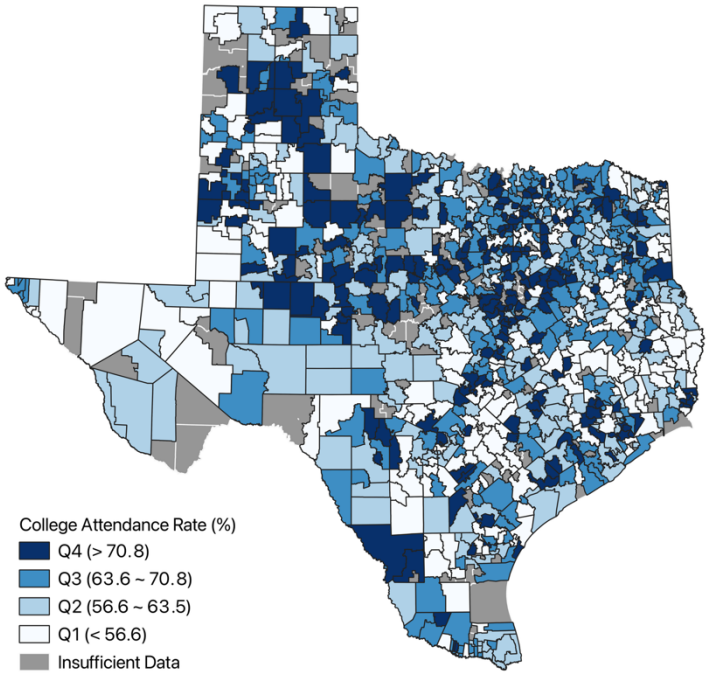


Figure 1. College Attendance Rates of Non-Movers by Texas School District, 1994–1999 Kindergarten Cohorts

The map shows the share of non-movers in each Texas school district who attend college within four years of expected high school graduation. Non-movers are students in the 1994 to 1999 kindergarten cohorts who never change school districts between kindergarten and grade 12. The map pools the six cohorts; the regressions use district opportunity computed separately for each cohort. Districts are grouped into quartiles of the distribution across districts, without weighting by enrollment. Districts with fewer than 20 non-movers are shown in gray to protect student privacy.

These differences, however, may reflect either the effect of districts or the sorting of families across districts (Katz 2026).[5] For example, families with more resources may choose districts with higher attainment, so part of a district's high mean reflects who lives there rather than what the district provides. To separate the two, I study students who move and identify the effect of districts from the timing of their moves (Chetty and Hendren 2018a, 2018b).

[5] The Opportunity Atlas, which reports the adult outcomes of children by the census tract in which they grew up for nearly the entire U.S. population, similarly measures opportunity based on children's actual outcomes (Chetty et al. 2026). Using these data, Chetty et al. (2026) find that about 60% of the variation in upward mobility across census tracts reflects causal place effects and the rest reflects sorting (see also Katz 2026).

For each one-time mover, the treatment is the difference in district opportunity between the destination and origin districts for the student's own kindergarten cohort, which I denote Δ Opportunity. On average, one-time movers move to higher-opportunity districts. The mean Δ Opportunity for college attendance is 4.19 percentage points, with a standard deviation of 13.05 percentage points (Table A2). The standard deviation is three times the mean, so the sample includes many moves to lower-opportunity districts as well as to higher-opportunity ones. The distribution is similar for SWDs (mean 3.47, standard deviation 12.90) and students without disabilities (mean 4.33, standard deviation 13.08). The two groups therefore face similar variation in the treatment, which allows me to compare their exposure effects directly.

4. Empirical Strategy

4.1 Childhood Exposure Design

I estimate childhood exposure effects by comparing one-time movers who move at different ages to districts of different opportunity (Chetty and Hendren 2018a). If places affect educational attainment, a student who moves to a higher-opportunity district earlier spends more years there and should gain more. Let Y_i denote an outcome for student i, who belongs to kindergarten cohort $c(i)$ and moves from origin district $o(i)$ to destination district $d(i)$ in grade m_i. In the equations, I write $\Delta Opp_i = Opp_{d(i)c(i)} - Opp_{o(i)c(i)}$ for Δ Opportunity, where Opp_{dc} is district opportunity in district d for cohort c, and $Dur_i = 13 - m_i$ for Duration, the number of years the student spends in the destination. I estimate

$$Y_i = \alpha_{o(i)c(i)} + \sum_{e=1}^{12} \theta_e \mathbf{1}\{Dur_i = e\} + \beta\, \Delta Opp_i + \gamma\, (\Delta Opp_i \times Dur_i) + X_i'\lambda + \varepsilon_i \qquad (1)$$

where $\alpha_{o(i)c(i)}$ are cohort-by-origin-district fixed effects, θ_e are fixed effects for each duration of exposure, and X_i contains indicators for sex and race or ethnicity. For each outcome, ΔOpp_i is measured with district opportunity for the same outcome.

The coefficient of interest is γ. It measures how much of the origin-destination gap in district opportunity a student closes for each additional year in the destination. For example, $\gamma = 0.011$ implies that a student who moves from a district where 50% of non-movers attend college to a district where 70% attend raises her probability of attending college by 0.22 percentage points, or 1.1% of the 20-point gap, for each additional year in the destination. A student who moves in grade 1 rather than grade 12 spends 11 more years in the destination by high school graduation and gains about 2.4 percentage points more. The coefficient β captures the part of the relationship between ΔOpp_i and the outcome that does not grow with years in the destination. It reflects mainly selection, the tendency of families who move to higher-opportunity districts to have children with better outcomes regardless of when they move.

Each set of fixed effects has a purpose. The cohort-by-origin fixed effects compare students who start in the same district in the same cohort, so differences in origin opportunity and cohort-specific shocks do not affect the estimates. The duration fixed effects absorb any effect of moving in a given grade that is common to all destinations, such as the disruption of changing schools (Hanushek et al. 2004; South et al. 2007). I cluster standard errors by origin district, because students from the same origin share the fixed effect and the origin component of ΔOpp_i.

To compare students with and without disabilities, I interact ΔOpp_i, $\Delta Opp_i \times Dur_i$, and the duration fixed effects with SWD_i, an indicator for disability status in grade 5, and estimate

$$Y_i = \alpha_{o(i)c(i)} + \sum_{e=1}^{12} (\theta_e + \theta_e^D SWD_i)\mathbf{1}\{Dur_i = e\} + \beta\, \Delta Opp_i + \beta^D(\Delta Opp_i \times SWD_i)$$

$$+ \gamma\, (\Delta Opp_i \times Dur_i)\ + \gamma^D(\Delta Opp_i \times Dur_i \times SWD_i) + X_i'\lambda + \varepsilon_i \qquad (2)$$

The exposure effect is γ for students without disabilities and $\gamma + \gamma^D$ for SWDs, and γ^D measures the difference between the two groups.

4.2 Exposure Profiles by Grade at Move

Equations (1) and (2) assume that each year of exposure has the same effect. To examine the timing of exposure, I replace the linear duration term with separate coefficients for each grade at move and estimate

$$Y_i = \alpha_{o(i)c(i)} + \sum_{m=1}^{12} \mathbf{1}\{m_i = m\}(\theta_m + \theta_m^D SWD_i)$$

$$+ \sum_{m=1}^{12} \mathbf{1}\{m_i = m\}(b_m\, \Delta Opp_i + b_m^D\, \Delta Opp_i \times SWD_i) + X_i'\lambda + \varepsilon_i \qquad (3)$$

The coefficient b_m measures how strongly the outcomes of students without disabilities who move in grade m track ΔOpp_i, and $b_m + b_m^D$ does the same for SWDs. Figure 2 plots these coefficients.

Each b_m combines two components. The first is selection, the tendency of families who move to higher-opportunity districts to have children with better outcomes. The second is exposure, which grows with the years remaining in the destination. If selection does not vary with the grade at move and each year of exposure has the same effect, $b_m = \delta + \gamma(13 - m)$, so the level of the profile reflects selection and its slope reflects exposure (Chetty and Hendren 2018a). I summarize

each profile by the slope of an equal-weighted least squares line through the 12 coefficients. I also test whether each profile is flat, whether the profiles differ by disability status, and whether they depart from a straight line (Table C1).

4.3 Identifying Assumption and Threats

The identifying assumption is that, among students from the same origin district and cohort, the relationship between ΔOpp_i and unobserved determinants of attainment does not vary with the grade at move (Chetty and Hendren 2018a). Families who move to higher-opportunity districts may differ from other families. The design requires only that this difference be the same for families who move when their child is in grade 2 and for families who move when their child is in grade 10. Three concerns deserve attention.

First, families may choose when to move based on their child's achievement. A family might move earlier if a child is doing well in school, or might delay a move until a child with a disability is settled in a set of services. If such timing is related to ΔOpp_i, the estimate of γ would be biased. To test for this, I estimate Equations (1) and (2) with grade 3 test scores as the outcome among students who move after grade 3, whose scores are measured before the move. Under the identifying assumption, the coefficient on $\Delta Opp_i \times Dur_i$ should be zero. Section 6.1 reports these placebo tests.

Second, families who move between some pairs of districts may tend to move earlier than others, and these pairs may differ in ways that ΔOpp_i does not capture. To address this concern, I replace the cohort-by-origin fixed effects $\alpha_{o(i)c(i)}$ with cohort-by-origin-by-destination fixed effects $\alpha_{o(i)d(i)c(i)}$. This specification compares students who make the same move in the same

cohort but in different grades, so it identifies exposure effects only from the timing of moves (Chetty and Hendren 2018b). I also add controls for free or reduced-price lunch and English learner status and cluster standard errors by both origin and destination district. Section 6.2 reports these results.

Third, disability status may respond to the destination. For students who move before grade 5, disability status in grade 5 is measured after the move. The destination can affect this status in two ways. Districts differ in how they identify disabilities, and the destination may affect whether a disability emerges or becomes apparent in the elementary grades. Table B3 shows that each additional year in a district with 10 percentage points higher college attendance raises the probability of identification in grade 5 by 0.15 percentage points, while the corresponding coefficient is close to zero among students who move after grade 5. Among early movers, a comparison of SWDs and students without disabilities could therefore reflect who becomes an SWD rather than how districts affect SWDs. I address this concern in two ways. Section 5.3 estimates Equation (2) separately for students who move in grades 6 to 12, whose status is determined before the move, and Section 6.3 measures disability status in grades 1 to 4 instead of grade 5.[6]

5. Results

5.1 Average Exposure Effects

[6] I used Claude Opus 5.5 (Anthropic) for editorial support in preparing this manuscript. I used it to suggest sentence-level edits for clarity and style, check that the statistics reported in the text match those in the tables, and check the format of the reference list. The theoretical contributions, analytical choices, and arguments are my own, and I am fully responsible for the content of the article.

Table 2 reports the effects of childhood exposure to district opportunity on high school completion, college attendance, and college completion. Panel A shows that each additional year spent in a higher-opportunity district raises all three outcomes. The exposure effects are 0.011 for high school completion, 0.011 for college attendance, and 0.009 for college completion. Take college attendance as an example. Spending one more year in a district where non-movers' college attendance rate is 1 percentage point higher raises a mover's college attendance rate by 0.011 percentage points. That is, movers' college attendance converges to that of non-movers in the destination at a rate of 1.1% per year of exposure. A student who spends 12 years in the destination, from grade 1 through grade 12, therefore closes about 13% (12 × 1.1%) of the difference in non-movers' college attendance between the origin and the destination. The other two outcomes converge at similar rates, 1.1% per year for high school completion and 0.9% per year for college completion.

In percentage points, the size of these effects depends on how much district opportunity varies for each outcome. I therefore scale each estimate by the standard deviation of district opportunity for that outcome and by the average exposure of 7.55 years (Table A2). For high school completion, a one standard deviation increase in district opportunity raises the completion rate by 0.5 percentage points, 0.6% of the mean. This effect is small because high school completion varies less across districts (standard deviation 6.20 percentage points) and is already high among movers (86.3%). For college attendance, the same increase raises the attendance rate by 1.1 percentage points, 1.9% of the mean. For college completion, it raises the completion rate by 0.7 percentage points, 4.5% of the mean. Relative to the mean, the gain is largest for college completion, the least common outcome among movers.

Panel B reports the exposure effects separately for SWDs and students without disabilities. For students without disabilities, the exposure effects are 0.012 for high school completion, 0.012 for college attendance, and 0.011 for college completion. For SWDs, they are 0.010, 0.024, and 0.009. The two groups differ only in college attendance. The difference of 0.012 is statistically significant at the 10% level, and the college attendance of SWDs converges to that of non-movers in the destination at 2.4% per year of exposure, twice the rate for students without disabilities. For high school completion and college completion, the differences are small and not statistically significant. The estimate for SWDs on high school completion is also imprecise.

As a share of each group's mean, the gains for SWDs are larger than those for students without disabilities on both college outcomes. For college attendance, the gain is 2.3 percentage points for SWDs, compared with 1.2 percentage points for students without disabilities. Because SWDs also attend college at lower rates (35.3%, compared with 60.3%), these gains are 6.7% and 2.0% of the respective means. For college completion, the gain is similar in percentage points, 0.7 for SWDs and 0.8 for students without disabilities. Only 5.1% of SWDs complete college, however, compared with 17.6% of students without disabilities. The gain for SWDs is 13% of the mean, compared with 4.8% for students without disabilities.

Table 2. Childhood Exposure Effects on Educational Attainment

	High school completion (1)	College attendance (2)	College completion (3)
A. All students			
Exposure effect	0.011***	0.011***	0.009***
	(0.003)	(0.004)	(0.003)
B. By disability status			
Without disabilities	0.012***	0.012***	0.011***
	(0.003)	(0.005)	(0.004)
With disabilities	0.010	0.024***	0.009***
	(0.011)	(0.006)	(0.003)
Difference (with minus without)	-0.003	0.012*	-0.002
	(0.011)	(0.007)	(0.005)
Observations	308,286	272,414	272,414

Notes: Each estimate is the coefficient on Δ Opportunity × Duration from a regression of the outcome on Δ Opportunity, Δ Opportunity × Duration, and indicators for Duration. District opportunity is the mean of the outcome among non-movers (students who never change districts between kindergarten and grade 12) in the same kindergarten cohort. Δ Opportunity is the destination minus origin difference in district opportunity for the outcome in each column, and Duration is the number of years of exposure to the destination district (13 minus grade at move). Panel B interacts every term with an indicator for disability status; the difference is the coefficient on the three-way interaction. A coefficient of 0.011 means that each additional year in the destination closes 1.1% of the gap in district opportunity between the origin and destination. The number of observations is smaller than in Table 1 because of masked district opportunity, missing outcomes, and the removal of singleton fixed-effect cells. All regressions include cohort-by-origin-district fixed effects and controls for sex and race or ethnicity. Standard errors clustered by origin district are in parentheses. *p < 0.10; **p < 0.05; ***p < 0.01.

5.2 Exposure Effects by Grade at Move

Figure 2 plots the coefficients on Δ Opportunity by the grade in which the student moves, estimated from Equation (3). For students without disabilities, the coefficients show two patterns. First, they decline with the grade at move for all three outcomes. Students who move to a higher-opportunity district earlier, and therefore spend more years there, gain more. The fitted lines fall by 0.009 per grade for high school completion, 0.009 for college attendance, and 0.007 for college completion, so each additional year of exposure raises the coefficient by these amounts, somewhat

less than the estimates in Table 2.[7] These slopes are statistically significant at the 5% level for high school completion and at the 10% level for the two college outcomes, and for each outcome the test that the 12 coefficients are equal rejects at the 1% level (Table C1). Second, the coefficients remain between about 0.45 and 0.60 even for students who move in grade 12. Because these students spend only one year in the destination, exposure can explain little of this level. Under the identifying assumption in Section 4.3, nearly all of it reflects selection, the tendency of families who move to higher-opportunity districts to have children with better outcomes.

[7] The slopes in Figure 2 weight each grade at move equally, while the regression in Table 2 weights each student, so the two sets of estimates differ somewhat (Chetty and Hendren 2018a, 1136n27).

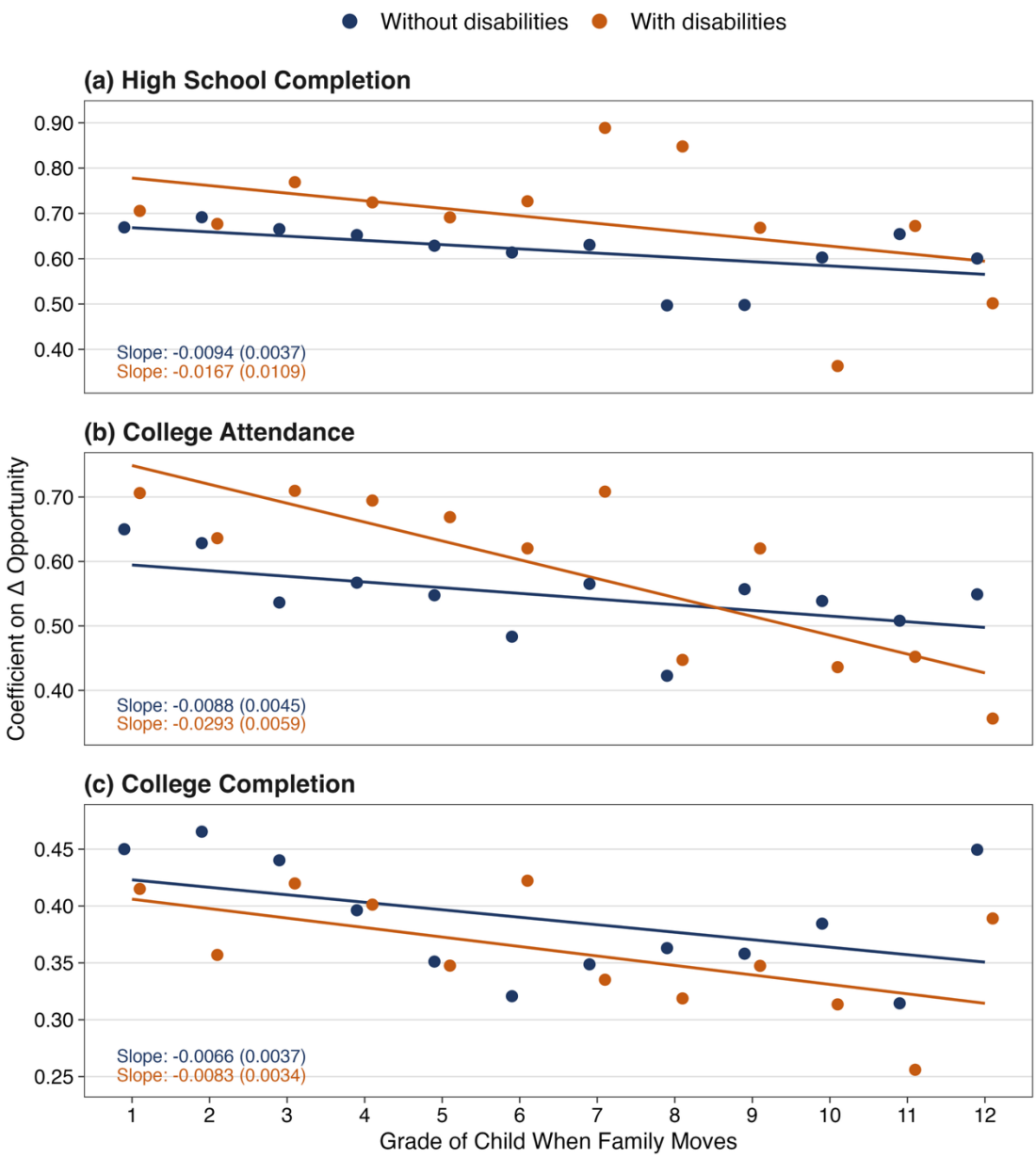


Figure 2. Childhood Exposure Effects on Educational Attainment by Grade at Move

Each point is the coefficient on Δ Opportunity for students who move in the indicated grade, estimated from Equation (3), which interacts Δ Opportunity with indicators for grade at move and with disability status. The regression includes cohort-by-origin-district fixed effects and controls for sex and race or ethnicity. Each line is a least squares fit of the 12 coefficients on the grade at move, with each grade weighted equally. The reported slope is the change in the coefficient per grade at move, with its standard error clustered by origin district in parentheses. Its magnitude equals the exposure effect per additional year of exposure in Table C1. The height of each line also reflects selection (Section 4.2).

For college attendance, the coefficients for SWDs decline much faster. The fitted line for SWDs falls by 0.029 per grade, more than three times the 0.009 for students without disabilities, and the difference of 0.020 is statistically significant at the 1% level (Table C1). Could this steeper decline reflect stronger selection rather than larger exposure effects? If the families of SWDs who move to higher-opportunity districts were more advantaged than other moving families, the coefficients for SWDs would be higher in every grade, including late grades in which exposure is short.[8] They are not. The two lines cross between grades 8 and 9, and for moves in grades 10 to 12, which leave at most three years in the destination, the coefficient for SWDs is below that for students without disabilities in every grade. The larger coefficients for SWDs appear only among moves in grades 1 to 7. This pattern suggests that SWDs gain more from moves before grade 8 because of longer exposure, not because their families select into destinations more strongly.

For the other two outcomes, the slopes do not differ significantly by disability status (Table C1). For high school completion, the line for SWDs falls faster, by 0.017 per grade, but the coefficients for SWDs vary widely across grades, from about 0.37 in grade 10 to about 0.89 in grade 7, and the slope is imprecisely estimated. For college completion, the two lines are nearly parallel, falling by 0.007 per grade for students without disabilities and 0.008 for SWDs.

The coefficients do not follow a straight line exactly. For college attendance, both the quadratic component for students without disabilities and its difference between the two groups are statistically significant at the 1% level (Table C1). For students without disabilities, the college attendance and college completion coefficients fall mainly between grades 1 and 6 and show no further decline afterward. For SWDs, the college attendance coefficients stay between about 0.62

[8] In these grades, the coefficients mostly reflect selection. For SWDs who move in grade 12, one year of exposure accounts for less than 10% of the fitted coefficient for college attendance (0.029 compared with about 0.43).

and 0.71 for moves in grades 1 to 7 and fall to between about 0.36 and 0.45 in grades 8, 10, 11, and 12. The slopes therefore summarize the average change across grades rather than a constant effect of each year.

5.3 Moves Before and After Grade 5

I measure disability status in grade 5 because many SWDs are not identified until the later elementary grades (Section 3.2). For students who move before grade 5, however, this status is measured after the move and may respond to the destination (Section 4.3), so the comparison in Panel B of Table 2 could reflect who becomes an SWD rather than how districts affect SWDs. Table 3 addresses this concern by splitting the sample at grade 5. Panel A includes students who move in grades 1 to 5, who make up 58% of the estimation sample, and pools SWDs and students without disabilities because their disability status may depend on the destination. Panels B and C include students who move in grades 6 to 12, whose disability status is measured before the move. Within each panel, the exposure effect comes from comparing students who move in different grades within the same range.

For the average student, exposure effects are concentrated in moves before grade 6. Among students who move in grades 1 to 5, the exposure effects are 0.022 for college attendance and 0.021 for college completion, about twice the estimates for all movers in Table 2 (Panel A). Among students who move in grades 6 to 12, the corresponding estimates are 0.003 and -0.002 and are not statistically significant (Panel B). For high school completion, the estimate is 0.009 in both ranges and is not statistically significant in either. This pattern matches Figure 2, where the coefficients for students without disabilities fall most steeply among early moves. It is also consistent with

evidence from the Moving to Opportunity experiment, in which moving to a lower-poverty neighborhood before age 13 increased college attendance and earnings, while moving as an adolescent had slightly negative effects (Chetty et al. 2016).

The near-zero estimates for late movers, however, mask a large difference by disability status. For students without disabilities who move in grades 6 to 12, the exposure effects are close to zero for all three outcomes, consistent with their flat coefficients after grade 6 in Figure 2. For SWDs, they are positive and statistically significant for all three outcomes, 0.056 for high school completion, 0.044 for college attendance, and 0.019 for college completion (Panel C). The differences between the two groups are statistically significant for all three outcomes, although only marginally for high school completion. They are also larger than in Table 2, which pools early and late movers. For college attendance, the difference is 0.052, compared with 0.012 in Table 2, and for high school completion and college completion, the differences are not statistically significant in Table 2.

These effects are large. Among SWDs who move in grades 6 to 12, college attendance converges to that of non-movers in the destination at 4.4% per year of exposure. An SWD who moves in grade 6 rather than grade 12 to a district with one standard deviation higher opportunity is 3.4 percentage points more likely to attend college, about 10% of the mean college attendance rate of SWDs (35.3%).

Because the differences are larger among students whose disability status is fixed before the move, the larger exposure effects for SWDs in Table 2 are unlikely to arise from the response of disability status to the destination. Panel C also shows that the timing of exposure differs by disability status. For students without disabilities, the gains from a higher-opportunity district come mainly from years spent there in the elementary grades. For SWDs, years spent there in

middle and high school also raise attainment, consistent with the sharp decline in their college attendance coefficients after grade 7 in Figure 2. Panel A cannot show whether SWDs also gain more from moves in grades 1 to 5, because disability status for these students may depend on the destination. Section 6.3 examines this question by measuring disability status in grades 1 to 4 (Tables B4 and B5).

Table 3. Childhood Exposure Effects on Educational Attainment for Moves Before and After Grade 5

	High school completion (1)	College attendance (2)	College completion (3)
A. Movers in grades 1 to 5, all students			
Exposure effect	0.009	0.022**	0.021**
	(0.009)	(0.009)	(0.009)
Observations	179,454	159,010	159,010
B. Movers in grades 6 to 12, all students			
Exposure effect	0.009	0.003	-0.002
	(0.009)	(0.007)	(0.007)
Observations	128,056	112,624	112,624
C. Movers in grades 6 to 12, by disability status			
Without disabilities	0.000	-0.008	-0.008
	(0.010)	(0.008)	(0.008)
With disabilities	0.056**	0.044***	0.019**
	(0.028)	(0.014)	(0.010)
Difference (with minus without)	0.057*	0.052***	0.027**
	(0.030)	(0.014)	(0.011)
Observations	128,056	112,624	112,624

Notes: Specifications follow Table 2. Panel A includes students who move in grades 1 to 5, and Panels B and C include students who move in grades 6 to 12. For students who move in grades 1 to 5, disability identification in grade 5 occurs after the move and responds to destination opportunity (Table B3). Panel A therefore reports estimates for all students only, and no disability indicator enters that specification. In Panel C, disability status is determined before the move for every student. All regressions include cohort-by-origin-district fixed effects and controls for sex and race or ethnicity. Standard errors clustered by origin district are in parentheses. *p < 0.10; **p < 0.05; ***p < 0.01.

5.4 Effects by Disability Type

SWDs are a heterogeneous group, so the larger exposure effects in Table 2 may come from one type of disability. Learning disabilities alone account for 63.9% of SWDs (Table 1). Table 4 estimates separate exposure effects for students with learning disabilities and students with other disabilities. Panel A uses all one-time movers, and Panel B uses students who move after grade 5, whose disability type is measured in grade 5, before the move.

Among all movers (Panel A), both types of SWDs have larger exposure effects on college attendance than students without disabilities. The effects are 0.025 for students with learning disabilities and 0.022 for students with other disabilities, compared with 0.012 for students without disabilities, although only the difference for learning disabilities is statistically significant. For college completion, the effect is 0.015 for students with learning disabilities, slightly above the 0.011 for students without disabilities, but -0.002 for students with other disabilities. For high school completion, the effects for both types are close to that for students without disabilities (0.010 and 0.008, compared with 0.012).

Among students who move after grade 5 (Panel B), the results by type are consistent with those in Table 3, Panel C. The exposure effects for students without disabilities are close to zero, while the estimates for both types of SWDs are positive for all three outcomes. For college attendance, the effects are 0.033 for students with learning disabilities and 0.065 for students with other disabilities, and both are significantly larger than the effect for students without disabilities. For college completion, the effects are 0.010 and 0.033, compared with -0.008 for students without disabilities, and only the difference for other disabilities is statistically significant. For high school completion, the effects are 0.056 for both types but are imprecisely estimated. The estimates for

other disabilities are larger for college attendance and college completion, but the estimates for both types in this sample are imprecise.

The larger exposure effects for SWDs are not specific to learning disabilities. For college attendance, students with other disabilities have exposure effects similar to or larger than those of students with learning disabilities in both panels, and among students who move after grade 5, both types gain more than students without disabilities. For college completion, the estimates by type differ between the two panels and should be interpreted with caution, because Panel A includes students whose disability type is measured after the move (Section 5.3).

Table 4. Childhood Exposure Effects on Educational Attainment by Disability Type

	High school completion (1)	College attendance (2)	College completion (3)
A. All movers			
Exposure effects on			
Without disabilities	0.012***	0.012***	0.011***
	(0.003)	(0.005)	(0.004)
Learning disabilities	0.010	0.025***	0.015***
	(0.015)	(0.007)	(0.004)
Other disabilities	0.008	0.022**	-0.002
	(0.013)	(0.009)	(0.005)
Difference from students without disabilities			
Learning disabilities	-0.002	0.013*	0.004
	(0.015)	(0.008)	(0.006)
Other disabilities	-0.004	0.009	-0.013*
	(0.013)	(0.011)	(0.007)
Observations	308,286	272,414	272,414
B. Movers in grades 6 to 12			
Exposure effects on			
Without disabilities	0.000	-0.008	-0.008
	(0.010)	(0.008)	(0.008)
Learning disabilities	0.056	0.033	0.010
	(0.041)	(0.020)	(0.011)
Other disabilities	0.056	0.065***	0.033**
	(0.036)	(0.025)	(0.016)
Difference from students without disabilities			
Learning disabilities	0.057	0.040**	0.019
	(0.043)	(0.019)	(0.013)
Other disabilities	0.057	0.073***	0.041**
	(0.037)	(0.027)	(0.016)
Observations	128,056	112,624	112,624

Notes: Specifications follow Panel B of Table 2, with disability status split into learning disabilities and other disabilities, each interacted with every term. Panel A includes all one-time movers, and Panel B includes students who move in grades 6 to 12, whose disability type is determined before the move. Learning disabilities are specific learning disabilities, and other disabilities include all remaining categories. Differences are relative to students without disabilities. All regressions include cohort-by-origin-district fixed effects and controls for sex and race or ethnicity. Standard errors clustered by origin district are in parentheses. *p < 0.10; **p < 0.05; ***p < 0.01.

6. Validity and Robustness

The estimates in Section 5 rely on the assumption that the relationship between district opportunity and unobserved determinants of attainment does not vary with the grade at move (Section 4.3). The comparison of SWDs and students without disabilities also requires that disability status be determined independently of the destination district. This section examines three concerns about these assumptions, which involve the timing of moves, differences across origin-destination pairs, and the measurement of disability status.

6.1 Timing of Moves and Prior Achievement

If families with higher-achieving children move earlier to higher-opportunity districts, students with longer exposure would have higher achievement before the move, and the exposure effects would partly reflect prior achievement rather than the effects of districts. To test for this, I restrict the sample to students who move after grade 3 and estimate Equations (1) and (2) with grade 3 reading and math scores as a placebo outcome. Because these scores are measured before the move for this sample, the destination district cannot affect them, and the exposure effects should be zero under the identifying assumption. When I pool SWDs and students without disabilities, the estimates are small and not statistically significant for both subjects and all three measures of district opportunity (Table C2).

When I estimate the exposure effects separately by disability status, most estimates are not statistically significant. Of the six estimates for students without disabilities, only the one for the math score with district opportunity measured by high school completion is marginally significant. For SWDs, only the estimate for the math score with district opportunity measured by college

completion is statistically significant, and the difference between the two groups for the same estimate is the only one that is marginally significant. The estimates for SWDs are less precise, because only 41% of SWDs have a grade 3 score, compared with 89% of students without disabilities, as many students in special education in Texas took modified tests or were not tested in this period (Hanushek and Rivkin 2009; Ballis and Heath 2021).

To gauge the size of these estimates, I multiply each by the association between grade 3 scores and the corresponding outcome (Table C3).[9] The product is the bias in the exposure effect that differences in prior achievement would create. For the pooled estimates, the implied bias is at most 0.001, less than 10% of the exposure effects in Table 2. The two marginally significant estimates imply biases of 0.002 in the exposure effect on high school completion for students without disabilities (0.034 × 0.057) and 0.004 in the difference in college completion (0.075 × 0.053). The latter would make the difference in college completion, which is close to zero in Table 2 (-0.002), appear larger than it is. The significant estimate for SWDs implies a bias of 0.005 (0.098 × 0.053) in their exposure effect on college completion, about half of that effect in Table 2 (0.009), so the estimate for college completion among SWDs should be interpreted with caution. For college attendance, the only outcome with a statistically significant difference in Table 2, the implied bias in the difference is at most 0.002, about one-sixth of that difference. Differences in prior achievement therefore explain little of the exposure effects or of the larger effect on college attendance for SWDs, with the possible exception of the effect on college completion among SWDs.

[9] This calculation follows the omitted variable bias formula. The placebo estimate shows how much higher grade 3 scores are for each additional year of exposure to a higher-opportunity district, and Table C3 shows how much higher each outcome is for each standard deviation of grade 3 scores. Their product is the part of the exposure effect that differences in prior achievement would explain, in the same units as the exposure effects in Table 2.

6.2 Differences Across Origin-Destination Pairs

The cohort-by-origin fixed effects compare students who leave the same district, but families who move to different destinations may differ in ways that district opportunity does not capture. If families who move between some pairs of districts also tend to move earlier than others, these differences could bias the exposure effects. To address this concern, I replace cohort-by-origin fixed effects with cohort-by-origin-by-destination fixed effects (Table C4, Panel B). The exposure effects are then identified only from differences in the grade at move among students who make the same move in the same cohort, rather than from where families move (Chetty and Hendren 2018b).

In this specification, the exposure effects on college attendance and college completion remain positive and statistically significant for both groups. The difference in college attendance between SWDs and students without disabilities, however, falls by half, from 0.012 to 0.006, and is no longer statistically significant, because the estimate for students without disabilities rises from 0.012 to 0.015 while the estimate for SWDs falls from 0.024 to 0.020. For high school completion, the difference becomes negative and statistically significant. This specification relies on much less variation, because it uses only differences in the grade at move among students who make the same move in the same cohort, and it drops 52,120 students who are the only student making their move in their cohort. Adding controls for free or reduced-price lunch and English learner status changes the estimates little, with either set of fixed effects (Panels A and C). Clustering by both origin and destination district also leaves the standard errors largely unchanged (Panel D).

Among students who move after grade 5, the difference in college attendance is more robust (Table C5). Adding controls changes it little (Panel A). With cohort-by-origin-by-destination fixed effects, it falls from 0.052 to 0.032 but remains statistically significant (Panel B), and it is 0.028 and marginally significant when the controls are also added (Panel C). With standard errors clustered by both origin and destination district, this estimate is statistically significant (Panel D). The differences in high school completion and college completion remain positive in all specifications but are less precise with these fixed effects, because the specification uses only variation within origin-destination pairs and drops about 23% of the sample. For college completion, the estimate changes little (from 0.027 to 0.021), but its standard error rises from 0.011 to 0.016, and it is no longer statistically significant. For high school completion, the estimate falls by about half (from 0.057 to 0.028) and is not statistically significant. Comparing only students who make the same move therefore leaves the exposure effects and the larger effect on college attendance for SWDs who move after grade 5 positive and statistically significant, but it reduces the larger effect among all movers by half.

6.3 Timing of Disability Identification

Because disability status in grade 5 may respond to the destination for students who move before grade 5 (Section 4.3), Section 5.3 estimates the effects separately for students who move after grade 5. Here I also measure disability status in grades 1 to 4 instead of grade 5.

For students who move after grade 5, status is determined before the move in every grade of measurement. Among these students, the differences between SWDs and students without disabilities are positive for all three outcomes and every grade of measurement (Table B5). They

are similar when status is measured in grade 4 (0.068 for high school completion, 0.049 for college attendance, and 0.026 for college completion). When status is measured in grades 1 to 3, the differences in college attendance and college completion remain positive but are smaller and not statistically significant, while the differences in high school completion vary more across grades of measurement. This decline is expected, because identification grows rapidly through elementary school and levels off by grade 5 (Ballis and Heath 2021), so earlier measures identify fewer SWDs and a different mix of disabilities.

Measuring status in grade 1 also places it before the move for students who move in grades 2 to 5. Among all movers, however, the difference in college attendance is close to zero with this measure (Table B4). This result is difficult to compare with the main estimates, because students identified in grade 1 differ from those identified in grade 5. Learning disabilities make up 63.9% of SWDs identified in grade 5 but only 13.3% of those identified in grade 1, and speech impairments make up 69.7% of those identified in grade 1 (Table B1). This check therefore cannot show whether SWDs as identified in grade 5 gain more from early moves. Among students who move after grade 5, the larger effects for SWDs remain positive whichever grade is used to measure disability status, but the larger effect among all movers may partly reflect identification after the move.

7. Discussion

This paper shows that the educational attainment of SWDs depends on where they grow up, and that the ages at which places matter differ by disability status. Among Texas students who move across school districts, each additional year in a higher-opportunity district raises high school completion, college attendance, and college completion. For students without disabilities, moves

after grade 5 have no detectable effect, so their gains come mainly from moves in the elementary grades. For SWDs, moves after grade 5 still raise attainment, and the evidence is clearest for college attendance. The larger effects on college attendance are not confined to students with learning disabilities, the most common type of disability.

The average exposure effects are smaller than those in prior studies. Movers' college attendance converges to that of non-movers in the destination at 1.1% per year, compared with 3.7% per year across commuting zones in the United States (Chetty and Hendren 2018a). One reason may be that school districts are much smaller than commuting zones, and exposure effects tend to be smaller across smaller places. Chetty and Hendren (2018a) find slightly smaller exposure effects on income across counties than across commuting zones, and in Montreal, Laliberté (2021) finds smaller exposure effects on educational attainment across census tracts than across larger neighborhoods. Within a labor market, families can move to a nearby place, such as a better school district, so differences in outcomes across nearby places may partly reflect which families choose to live there (Chetty and Hendren 2018a). This sorting does not bias the exposure effects, which are identified from how the gains from moving to a higher-opportunity district vary with the grade at move, but it means that a smaller share of the differences across districts reflects their causal effects. Movers' outcomes therefore converge more slowly toward district opportunity.

SWDs may continue to gain from moves in middle and high school because decisions that shape their paths after high school are made in these grades. Most SWDs remain in special education into high school, and as graduation approaches, their IEPs turn to transition plans that guide them toward college or toward employment (Ballis and Heath 2021). A move in these grades places an SWD in the district that will shape these plans, so moving to a higher-opportunity district

can still raise attainment. The results also suggest that, at least in this setting, the disruption of services and relationships that a move can cause does not outweigh these gains.

These results show that the ages at which places matter can differ across groups of children. For students without disabilities, who gain little from moves after grade 5, the results match evidence for children in general that the gains from better places accumulate over childhood and that moves in adolescence bring smaller gains (Chetty et al. 2016; Chyn 2018). Chetty and Hendren (2018a) find that each year of childhood matters roughly equally on average across all children, but my results suggest that such an average may combine different patterns for SWDs and other students. Deutscher (2020) similarly finds that the ages at which places matter most differ across groups of children, in his case by family income.

The gains for SWDs are clearest for college attendance. Several features of the outcomes may explain why. High school completion varies less across districts than college attendance and is already high among movers, which leaves less room for districts to raise it. For SWDs, high school completion may also partly reflect modified graduation requirements, under which students in special education may graduate without passing the exit exam required of other students (Ballis and Heath 2021). College completion, in contrast, is rare among SWDs (5.1%), so effects on it are estimated less precisely. College attendance has no such modification, varies widely across districts, and follows directly from the transition plans made in high school.

These results have implications for housing and education policy. Because the exposure design separates the effects of districts from the sorting of families across them, the results suggest that the places where SWDs grow up affect their attainment. For children in general, evidence of such effects supports two kinds of policy, helping families move to higher-opportunity places and

improving the places where they already live (Chetty and Hendren 2018a, 2018b). The results here suggest that both are relevant for SWDs.

Helping families move can benefit SWDs even in middle and high school, because moves in these grades still raise their attainment. Many housing mobility programs, however, limit eligibility by children's age. The Seattle experiment, for example, enrolled only families with at least one child younger than 15 (Bergman et al. 2024), and limits of this kind may exclude families of older SWDs for whom a move would still pay off. These families may make up a meaningful share of those eligible for housing assistance, because SWDs are more likely than other students to come from low-income families (Table 1). Mobility programs could extend eligibility and counseling to these families and give them information about the special education services in destination districts.

Improving the places where SWDs already live could also help. Special education is planned for each student through an IEP but delivered by local schools, so the services a student receives depend on where the student lives (Section 2.2). If the services and transition planning available in middle and high school are part of what makes a place high-opportunity for SWDs, improving them could complement helping families move.

This study has several limitations. First, among all movers, the larger effect on college attendance for SWDs is not robust to comparing only students who make the same move in the same cohort, so the evidence for a difference by disability status comes mainly from students who move after grade 5. Second, like Chetty and Hendren (2018a), I estimate effects for families who chose to move, and the effects for families who move with housing assistance may differ. Third, I observe college outcomes only at Texas institutions, although only 1.7% of students in special education enroll in college outside Texas within two years of high school graduation (Ballis and

Heath 2021). Finally, the effect of a district combines the effects of its schools, peers, neighborhoods, and special education services, and I cannot separate them.

Future research could examine why places matter for SWDs. Comparing the IEPs of SWDs before and after a move would show whether moving to a higher-opportunity district changes the services they receive or the goals of their transition plans. Another question is whether connections between SWDs and students without disabilities help explain the gains SWDs receive from better districts, just as connections between people with low and high socioeconomic status are strongly associated with upward mobility (Chetty et al. 2022). It would also be useful to know how families of SWDs choose where to move. If these families have little information about the special education and social work services in other districts, the counseling and information that mobility programs provide may matter more for them than for other families.

This study shows that the question of when places matter has more than one answer. For students without disabilities, the gains from a better place come mainly from the early grades, but for SWDs, places continue to shape attainment through middle and high school, especially college attendance. Because SWDs rely on services that local schools provide, where they grow up shapes not only their schooling but also the paths they take after high school. Policies that help families reach better places, or that improve the places where they already live, need not stop at the elementary grades. For SWDs, educational opportunity remains tied to place well into adolescence.

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Appendix

Appendix A. Sample and District Opportunity

Table A1. Summary Statistics by Mobility Status

	Overall	Non-movers	One-time movers	Multi-movers
Students	1,244,591	679,810	347,586	217,195
Outcomes				
High school completion	0.878	0.912	0.863	0.795
College attendance	0.575	0.623	0.557	0.458
College completion	0.177	0.220	0.154	0.086
Race and ethnicity				
Hispanic	0.431	0.456	0.431	0.355
White	0.395	0.404	0.376	0.401
Black	0.147	0.115	0.162	0.224
Asian	0.020	0.021	0.024	0.012
Native American	0.003	0.003	0.004	0.004
Two or more races	0.002	0.002	0.003	0.003
Sex				
Female	0.477	0.481	0.473	0.475
Male	0.523	0.519	0.527	0.525
Socioeconomic status				
Free or reduced-price lunch	0.450	0.426	0.456	0.513
English learner	0.034	0.037	0.035	0.022
Baseline test scores (SD)				
Grade 3 reading score	-0.037	0.045	-0.089	-0.225
Grade 3 math score	-0.032	0.068	-0.094	-0.265
Disability				
Without disabilities	0.838	0.853	0.833	0.799
With disabilities	0.162	0.147	0.167	0.201
Disability type				
Learning disability	0.102	0.091	0.107	0.126
Speech impairment	0.021	0.022	0.020	0.019
Other health impairment	0.016	0.015	0.017	0.020
Emotional disturbance	0.009	0.006	0.009	0.020
Intellectual disability	0.007	0.007	0.007	0.009
Auditory or visual impairment	0.002	0.002	0.003	0.002
Autism	0.002	0.002	0.002	0.002
Orthopedic impairment	0.002	0.002	0.002	0.001
Traumatic brain injury	0.0003	0.0002	0.0004	0.001

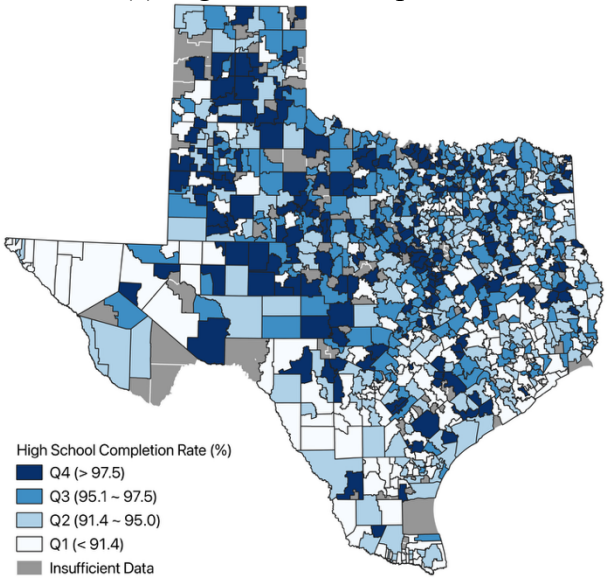
Notes: The sample is students in the 1994 to 1999 kindergarten cohorts in Texas public schools. Non-movers never change school districts between kindergarten and grade 12, one-time movers change districts once, and multi-movers change districts more than once. The first row reports the number of students. Test score rows report means, and all other rows report shares. Auditory or visual impairment includes deaf-blindness, which is combined with these categories to protect student privacy.

Table A2. Differences in District Opportunity Between Destination and Origin Districts

	High school completion	College attendance	College completion
All students	+2.13 (6.20)	+4.19 (13.05)	+3.32 (10.09)
Group mean (%)	86.3	55.7	15.4
Mean exposure duration (years)		7.55 (3.37)	
Without disabilities	+2.18 (6.19)	+4.33 (13.08)	+3.43 (10.11)
Group mean (%)	86.6	60.3	17.6
Mean exposure duration (years)		7.54 (3.36)	
With disabilities	+1.86 (6.26)	+3.47 (12.90)	+2.74 (9.97)
Group mean (%)	84.8	35.3	5.1
Mean exposure duration (years)		7.59 (3.47)	
Learning disabilities	+1.82 (6.25)	+3.42 (12.82)	+2.63 (9.83)
Group mean (%)	84.1	33.8	3.8
Mean exposure duration (years)		7.59 (3.48)	
Non-learning disabilities	+1.92 (6.28)	+3.55 (13.04)	+2.93 (10.20)
Group mean (%)	86.2	38.1	7.4
Mean exposure duration (years)		7.59 (3.44)	

Notes: For each group, the first row reports the mean difference in district opportunity between the destination and origin districts, in percentage points, with the standard deviation in parentheses. District opportunity for each outcome is the mean of that outcome among non-movers in the district in the student's kindergarten cohort. Group mean is the mean of the outcome in the group. Mean exposure duration is the number of years in the destination district (13 minus grade at move), with the standard deviation in parentheses.

(a) High School Completion



(b) College Completion

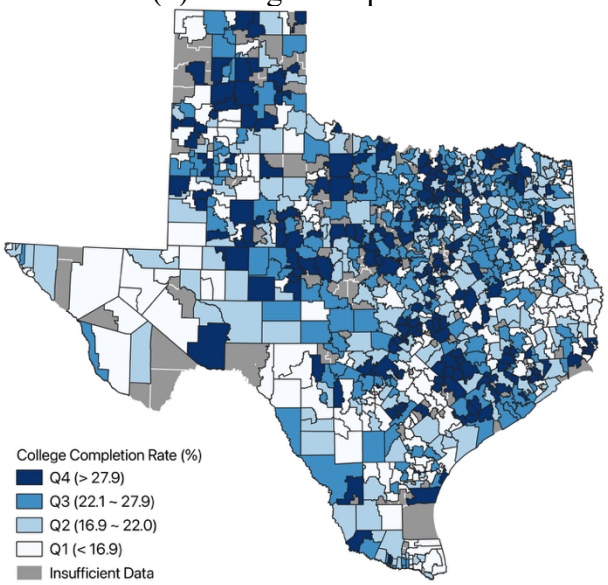


Figure A1. High School Completion and College Completion Rates of Non-Movers by Texas School District, 1994–1999 Kindergarten Cohorts

Panel (a) shows the share of non-movers in each Texas school district who complete high school, and Panel (b) shows the share who earn a postsecondary degree within six years of expected high school graduation. Non-movers are students in the 1994 to 1999 kindergarten cohorts who never change school districts between kindergarten and grade 12. The maps pool the six cohorts; the regressions use district opportunity computed separately for each cohort. Districts are grouped into quartiles of the distribution across districts, defined separately for each panel and without weighting by enrollment. Districts with fewer than 20 non-movers are shown in gray to protect student privacy.

Appendix B. Disability Status

Table B1. Disability Identification and Composition by Grade

Grade	Identified (%)	Learning disability (% of identified)	Speech impairment (% of identified)	Other (% of identified)
Kindergarten	5.3	4.7	78.1	17.2
1	8.1	13.3	69.7	17.0
2	10.9	33.2	47.4	19.4
3	13.1	47.6	31.6	20.8
4	15.6	59.0	19.3	21.7
5	16.7	63.9	12.1	24.0
6	16.7	67.4	6.8	25.8
7	16.2	68.6	3.9	27.5
8	15.8	69.5	2.3	28.2
9	15.2	69.9	1.2	28.9
10	14.7	69.6	0.7	29.7
11	14.3	69.6	0.5	29.9
12	13.7	69.9	0.4	29.7

Notes: Identified is the share of one-time movers observed in each grade who are identified with a disability in that grade. The remaining columns give the composition of identified students by primary disability. Other combines all remaining categories, including noncategorical early childhood, which appears only in kindergarten through grade 3. The number of students observed is 347,586 in kindergarten and grade 5 and falls to 310,796 in grade 12.

Table B2. One-Time Movers by Disability Status in Kindergarten and Grade 5

Kindergarten	Grade 5	N (%)	Grade 3 reading	Grade 3 math	High school completion	College attendance	College completion
Not identified	Not identified	281,846 (81.1%)	0.007	-0.002	0.866	0.603	0.177
Not identified	Identified	47,214 (13.6%)	-1.176	-0.968	0.834	0.355	0.049
Identified	Not identified	7,645 (2.2%)	-0.052	-0.024	0.874	0.594	0.169
Identified	Identified	10,881 (3.1%)	-0.729	-0.667	0.909	0.347	0.061

Notes: The table cross-classifies one-time movers by disability status in kindergarten and grade 5. N (%) is the number of students in each cell and its share of all one-time movers. Grade 3 test scores are standardized and are observed for fewer students in cells identified in grade 5; for example, scores are observed for 20,616 of the 47,214 students first identified after kindergarten. Outcome columns are shares.

Table B3. Destination District Opportunity and Disability Identification: Classification Response and Sorting

District opportunity measure	All one-time movers	One-time movers after grade 5
	(1)	(2)
A. Outcome: identified with a disability in grade 5		
High school completion	0.033***	0.005
	(0.006)	(0.011)
Observations	308,286	128,056
College attendance	0.015***	-0.004
	(0.004)	(0.005)
Observations	307,390	127,638
College completion	0.019***	-0.001
	(0.005)	(0.006)
Observations	307,390	127,638
B. Outcome: identified with a disability in kindergarten		
High school completion	0.006**	-0.003
	(0.003)	(0.008)
Observations	308,286	128,056
College attendance	0.002*	0.001
	(0.001)	(0.004)
Observations	307,390	127,638
College completion	0.004**	0.002
	(0.002)	(0.005)
Observations	307,390	127,638

Notes: Each estimate is the coefficient on Δ Opportunity × Duration from the specification in Panel A of Table 2, with an indicator for disability identification as the outcome and district opportunity measured by the outcome named in each row. In Panel A, column (2) restricts the sample to students who move after grade 5. Status for these students is determined before the move, so the coefficient is expected to be zero. In Panel B, status in kindergarten is determined before every move in the sample, so a nonzero coefficient indicates that students identified in kindergarten sort across destinations by the timing of the move. All regressions include cohort-by-origin-district fixed effects and controls for sex and race or ethnicity. Standard errors clustered by origin district are in parentheses. Coefficients labeled ***, **, and * are statistically significant at the 1%, 5%, and 10% levels, respectively.

Table B4. Childhood Exposure Effects by Grade of Disability Identification

Disability identification grade	Grade 5	Grade 4	Grade 3	Grade 2	Grade 1
	(1)	(2)	(3)	(4)	(5)
A. High school completion					
Without disabilities	0.012***	0.011***	0.010***	0.009***	0.010***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
With disabilities	0.010	0.008	0.017	0.012	0.017
	(0.011)	(0.012)	(0.010)	(0.012)	(0.012)
Difference (with minus without)	-0.003	-0.003	0.007	0.003	0.007
	(0.011)	(0.012)	(0.010)	(0.012)	(0.012)
Observations	308,286	300,846	298,284	297,150	299,213
B. College attendance					
Without disabilities	0.012***	0.011***	0.012***	0.011**	0.011***
	(0.005)	(0.004)	(0.004)	(0.004)	(0.004)
With disabilities	0.024***	0.024***	0.022***	0.021***	0.010
	(0.006)	(0.007)	(0.008)	(0.008)	(0.009)
Difference (with minus without)	0.012*	0.013*	0.011	0.010	-0.001
	(0.007)	(0.008)	(0.010)	(0.009)	(0.009)
Observations	272,414	265,836	263,482	262,466	264,325
C. College completion					
Without disabilities	0.011***	0.010***	0.010***	0.008***	0.008**
	(0.004)	(0.004)	(0.003)	(0.003)	(0.003)
With disabilities	0.009***	0.011***	0.013***	0.015***	0.015**
	(0.003)	(0.004)	(0.004)	(0.006)	(0.007)
Difference (with minus without)	-0.002	0.001	0.003	0.007	0.007
	(0.005)	(0.006)	(0.006)	(0.007)	(0.007)
Observations	272,414	265,836	263,482	262,466	264,325
Learning disabilities (% of identified)	63.9	59.0	47.6	33.2	13.3

Notes: Each column measures disability status in a different grade, from grade 5 in column (1) to grade 1 in column (5), and reports estimates from the specification in Panel B of Table 2. The sample is all one-time movers, so column (1) reproduces Panel B of Table 2. The last row reports the share of students identified with a disability in that grade whose primary disability is a learning disability (Table B1). For students who move before the grade in which disability status is measured, status is measured after the move; Table B5 repeats the analysis for students who move after grade 5, whose status in every column is determined before the move. The number of observations differs across columns because disability status is missing for some students in grades 1 to 4. All regressions include cohort-by-origin-district fixed effects and controls for sex and race or ethnicity. Standard errors clustered by origin district are in parentheses. Coefficients labeled ***, **, and * are statistically significant at the 1%, 5%, and 10% levels, respectively.

Table B5. Childhood Exposure Effects by Grade of Disability Identification (Movers After Grade 5)

Disability identification grade	Grade 5 (1)	Grade 4 (2)	Grade 3 (3)	Grade 2 (4)	Grade 1 (5)
A. High school completion					
Without disabilities	0.000	-0.004	0.005	0.005	0.002
	(0.010)	(0.011)	(0.010)	(0.010)	(0.010)
With disabilities	0.056**	0.064**	0.043*	0.036	0.074**
	(0.028)	(0.027)	(0.026)	(0.028)	(0.030)
Difference (with minus without)	0.057*	0.068**	0.038	0.031	0.072**
	(0.030)	(0.030)	(0.027)	(0.031)	(0.033)
Observations	128,056	125,958	125,453	125,241	125,743
B. College attendance					
Without disabilities	-0.008	-0.008	-0.002	0.000	-0.001
	(0.008)	(0.008)	(0.008)	(0.007)	(0.007)
With disabilities	0.044***	0.042***	0.017	0.024	0.030
	(0.014)	(0.016)	(0.016)	(0.018)	(0.023)
Difference (with minus without)	0.052***	0.049***	0.019	0.024	0.031
	(0.014)	(0.017)	(0.017)	(0.018)	(0.023)
Observations	112,624	110,778	110,277	110,075	110,566
C. College completion					
Without disabilities	-0.008	-0.008	-0.005	-0.005	-0.003
	(0.008)	(0.007)	(0.008)	(0.007)	(0.007)
With disabilities	0.019**	0.018*	0.013	0.013	0.010
	(0.010)	(0.010)	(0.011)	(0.013)	(0.015)
Difference (with minus without)	0.027**	0.026**	0.018	0.018	0.013
	(0.011)	(0.011)	(0.012)	(0.013)	(0.015)
Observations	112,624	110,778	110,277	110,075	110,566

Notes: The sample is students who move after grade 5. Each column measures disability status in a different grade, from grade 5 in column (1) to grade 1 in column (5), and reports estimates from the specification in Panel B of Table 2. Because every move occurs after grade 5, disability status in every column is determined before the move. Column (1) reproduces Panel C of Table 3. The number of observations differs across columns because disability status is missing for some students in grades 1 to 4. All regressions include cohort-by-origin-district fixed effects and controls for sex and race or ethnicity. Standard errors clustered by origin district are in parentheses. Coefficients labeled ***, **, and * are statistically significant at the 1%, 5%, and 10% levels, respectively.

Appendix C. Exposure Profiles and Robustness

Table C1. Slope and Shape of the Childhood Exposure Profile

	High school completion (1)	College attendance (2)	College completion (3)
A. Exposure slope (Δ Opportunity × Duration)			
Without disabilities	0.009**	0.009*	0.007*
	(0.004)	(0.004)	(0.004)
With disabilities	0.017	0.029***	0.008**
	(0.011)	(0.006)	(0.003)
Difference (with minus without)	0.007	0.020***	0.002
	(0.012)	(0.007)	(0.005)
Observations	308,286	272,414	272,414
B. Joint tests (F-statistic [p-value])			
Profile is flat, students without disabilities	2.98 [0.001]	3.42 [<0.001]	3.61 [<0.001]
Profile differs by disability status	1.21 [0.273]	2.27 [0.010]	1.39 [0.173]
Quadratic, students without disabilities	3.28 [0.070]	8.46 [0.004]	11.88 [0.001]
Quadratic, group difference	4.99 [0.026]	7.53 [0.006]	3.27 [0.071]

Notes: Panel A reports the exposure effect implied by each fitted line in Figure 2, the equal-weight least squares line through the 12 coefficients on Δ Opportunity by grade at move. Because exposure equals 13 minus the grade at move, the exposure effect equals the slope in Figure 2 with the sign reversed. For example, the slope of -0.029 for students with disabilities in college attendance corresponds to an exposure effect of 0.029. Because the exposure effect is a linear combination of coefficients from one regression, its standard error accounts for their covariance. Panel B reports F-statistics with p-values in brackets from the same regression. The first row tests whether the 12 coefficients for students without disabilities are equal, and the second tests whether the coefficients for students with disabilities differ from those for students without disabilities. The last two rows test the quadratic orthogonal polynomial component of the coefficients for students without disabilities and the difference in this component between the two groups. Each is a single linear combination, so its F-statistic equals the squared t-statistic. All regressions include cohort-by-origin-district fixed effects and controls for sex and race or ethnicity. Standard errors in Panel A, clustered by origin district, are in parentheses. *p < 0.10; **p < 0.05; ***p < 0.01.

Table C2. Placebo Test: Exposure Effects on Pre-Move Achievement

	Grade 3 reading score		Grade 3 math score	
	(1)	(2)	(3)	(4)
A. District opportunity measured by high school completion				
All students	0.012		0.007	
	(0.028)		(0.023)	
Without disabilities		0.028		0.034*
		(0.026)		(0.020)
With disabilities		0.078		0.046
		(0.099)		(0.078)
Difference (with minus without)		0.050		0.012
		(0.103)		(0.075)
Observations	150,554	150,554	152,816	152,816
B. District opportunity measured by college attendance				
All students	0.005		0.010	
	(0.018)		(0.013)	
Without disabilities		0.015		0.020
		(0.021)		(0.014)
With disabilities		0.036		0.049
		(0.039)		(0.037)
Difference (with minus without)		0.021		0.029
		(0.044)		(0.035)
Observations	150,079	150,079	152,338	152,338
C. District opportunity measured by college completion				
All students	0.011		0.015	
	(0.020)		(0.015)	
Without disabilities		0.020		0.023
		(0.022)		(0.016)
With disabilities		0.070		0.098**
		(0.052)		(0.049)
Difference (with minus without)		0.050		0.075*
		(0.057)		(0.045)
Observations	150,079	150,079	152,338	152,338

Notes: The outcome is the standardized grade 3 reading score (columns 1 and 2) or math score (columns 3 and 4), measured before the move. The sample is students who move after grade 3. Columns (1) and (3) follow Panel A of Table 2, and columns (2) and (4) follow Panel B. Each panel uses the district opportunity measure named in its heading. Under the identifying assumption, the coefficients are zero. All regressions include cohort-by-origin-district fixed effects and controls for sex and race or ethnicity. Standard errors clustered by origin district are in parentheses. *p < 0.10; **p < 0.05; ***p < 0.01.

Table C3. Association Between Grade 3 Achievement and Educational Attainment

	High school completion		College attendance		College completion	
	(1)	(2)	(3)	(4)	(5)	(6)
Grade 3 reading score	0.023***	0.021***	0.065***	0.058***	0.032***	0.029***
	(0.002)	(0.002)	(0.002)	(0.003)	(0.002)	(0.002)
Grade 3 math score	0.057***	0.059***	0.071***	0.064***	0.053***	0.049***
	(0.002)	(0.003)	(0.004)	(0.005)	(0.003)	(0.003)
Observations	148,719	114,401	130,061	100,413	130,061	100,413

Notes: Each column reports coefficients from a regression of the outcome on standardized grade 3 reading and math scores, with controls for sex and race or ethnicity. Odd columns include cohort-by-origin-district fixed effects, and even columns include cohort-by-origin-by-destination fixed effects. The sample is the placebo sample of Table C2 (students who move after grade 3 with unmasked district opportunity for the outcome), restricted to students with a non-missing outcome and both grade 3 scores. These coefficients convert the placebo estimates in Table C2 into the implied bias in the exposure effects (Section 6.1). Standard errors clustered by origin district are in parentheses. *p < 0.10; **p < 0.05; ***p < 0.01.

Table C4. Robustness to Alternative Controls, Fixed Effects, and Clustering

	High school completion (1)	College attendance (2)	College completion (3)
A. Additional controls for free or reduced-price lunch and English learner status			
Without disabilities	0.011***	0.010**	0.009**
	(0.003)	(0.005)	(0.003)
With disabilities	0.007	0.021***	0.006*
	(0.011)	(0.005)	(0.003)
Difference (with minus without)	-0.004	0.011*	-0.002
	(0.011)	(0.007)	(0.005)
Observations	308,286	272,414	272,414
B. Cohort-by-origin-by-destination fixed effects			
Without disabilities	0.011***	0.015**	0.013***
	(0.004)	(0.006)	(0.005)
With disabilities	-0.014	0.020***	0.012***
	(0.012)	(0.006)	(0.004)
Difference (with minus without)	-0.025**	0.006	-0.002
	(0.013)	(0.008)	(0.007)
Observations	256,304	227,073	227,073
C. Additional controls and cohort-by-origin-by-destination fixed effects			
Without disabilities	0.010**	0.012**	0.011**
	(0.004)	(0.006)	(0.004)
With disabilities	-0.017	0.016**	0.009*
	(0.012)	(0.006)	(0.004)
Difference (with minus without)	-0.027**	0.004	-0.002
	(0.012)	(0.008)	(0.006)
Observations	256,304	227,073	227,073
D. Panel C with standard errors clustered by origin and destination district			
Without disabilities	0.010***	0.012*	0.011**
	(0.003)	(0.006)	(0.005)
With disabilities	-0.017	0.016***	0.009**
	(0.012)	(0.006)	(0.004)
Difference (with minus without)	-0.027**	0.004	-0.002
	(0.012)	(0.008)	(0.006)
Observations	256,304	227,073	227,073

Notes: Specifications follow Panel B of Table 2. Panel A adds indicators for free or reduced-price lunch and English learner status, both measured in grade 5. Panel B replaces cohort-by-origin-district fixed effects with cohort-by-origin-by-destination fixed effects, which absorb the level of Δ Opportunity; dropping singleton cells removes 52,120 observations in column (1) and 45,544 in columns (2) and (3). Panel C combines the two changes. Panel D repeats Panel C with standard errors clustered two ways, by origin district and by destination district. All regressions include controls for sex and race or ethnicity. In Panels A to C, standard errors are clustered by origin district. Standard errors are in parentheses. *p < 0.10; **p < 0.05; ***p < 0.01.

Table C5. Robustness to Alternative Controls, Fixed Effects, and Clustering: Movers After Grade 5

	High school completion (1)	College attendance (2)	College completion (3)
A. Additional controls for free or reduced-price lunch and English learner status			
Without disabilities	-0.000	-0.008	-0.009
	(0.010)	(0.008)	(0.007)
With disabilities	0.055*	0.039***	0.016
	(0.028)	(0.014)	(0.010)
Difference (with minus without)	0.056*	0.047***	0.024**
	(0.030)	(0.014)	(0.011)
Observations	128,056	112,624	112,624
B. Cohort-by-origin-by-destination fixed effects			
Without disabilities	-0.003	0.002	-0.004
	(0.014)	(0.011)	(0.009)
With disabilities	0.025	0.034**	0.018
	(0.036)	(0.016)	(0.013)
Difference (with minus without)	0.028	0.032**	0.021
	(0.037)	(0.016)	(0.016)
Observations	98,323	86,815	86,815
C. Additional controls and cohort-by-origin-by-destination fixed effects			
Without disabilities	-0.001	0.002	-0.004
	(0.014)	(0.011)	(0.009)
With disabilities	0.023	0.030*	0.016
	(0.037)	(0.016)	(0.014)
Difference (with minus without)	0.025	0.028*	0.020
	(0.037)	(0.016)	(0.016)
Observations	98,323	86,815	86,815
D. Panel C with standard errors clustered by origin and destination district			
Without disabilities	-0.001	0.002	-0.004
	(0.016)	(0.011)	(0.009)
With disabilities	0.023	0.030*	0.016
	(0.038)	(0.017)	(0.013)
Difference (with minus without)	0.025	0.028**	0.020
	(0.036)	(0.014)	(0.015)
Observations	98,323	86,815	86,815

Notes: The sample is students who move in grades 6 to 12, whose disability status is determined before the move. Specifications follow Panel C of Table 3. Panel A adds indicators for free or reduced-price lunch and English learner status, both measured in grade 5. Panel B replaces cohort-by-origin-district fixed effects with cohort-by-origin-by-destination fixed effects, which absorb the level of Δ Opportunity; dropping singleton cells removes 30,240 observations in column (1) and 26,339 in columns (2) and (3). Panel C combines the two changes. Panel D repeats Panel C with standard errors clustered two ways, by origin district and by destination district. All regressions include controls for sex and race or ethnicity. In Panels A to C, standard errors are clustered by origin district. Standard errors are in parentheses. *p < 0.10; **p < 0.05; ***p < 0.01.